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Introduction

This report addresses three tasks from the [redacted] coursework: supervised classification of entity races using greyscale images, supervised regression to predict bribe amounts for human guards, and unsupervised clustering to group entities by physical threat level. Each task employs machine learning techniques covered in the unit, with results evaluated and visualized as required.

1 Question 1: Supervised Classification

1.1 Data Preparation

The dataset `sprites_greyscale10.npz` contains 10×10 greyscale images of entities labeled by race. Using NumPy, I loaded the training (`train_X`, `train_Y`) and test (`test_X`, `test_Y`) sets. I identified four unique classes and selected the first two (classes halfling and human) for binary classification. The filtered dataset retained 53.41% of the original training samples (5892 samples) and 63.31% of test samples (1734 samples) after reshaping into 100-dimensional feature vectors.

1.2 Model Training and Evaluation

Three classifiers—Decision Tree, K-Nearest Neighbors (KNN), and Logistic Regression—were implemented using scikit-learn:

- **Decision Tree**: Trained with default settings (no pruning, `max_depth=None`). Accuracy on the test set: 0.9377.
- **KNN**: Used `n_neighbors=5`. Accuracy: 0.9308.
- **Logistic Regression**: Default parameters (`C=1.0`, L2 regularization). Accuracy: 0.8668.

Decision Tree outperformed the others, possibly due to its ability to capture non-linear patterns in the image data. Logistic Regression's lower accuracy may stem from convergence issues (noted in the warning), though not addressed due to default parameter constraints.

1.3 Hyperparameter Justification

- **Decision Tree**: Default `max_depth=None` was chosen to allow full growth, balancing complexity and overfitting risk. Manual tuning (e.g., depths 5–15) could improve generalization but was not implemented per instructions.
- **KNN**: `n_neighbors=5` was selected as a moderate value, balancing local sensitivity and noise robustness, based on worksheet examples. Values like 1 or 10 were considered but not tested extensively.
- **Logistic Regression**: `C=1.0` (default regularization strength) was retained, as the dataset size and feature scale did not suggest strong overfitting or underfitting. Manual iteration over `C` (e.g., 0.1, 10) was avoided to comply with the no-optimization rule.

Decision Tree's superior accuracy justifies its use for this task.

2 Question 2: Supervised Regression

2.1 Data Preparation

The datasets `dungeon_sensorstats_train.csv` and `dungeon_sensorstats_test.csv` were loaded using pandas. I filtered for the 'human' race and extracted four features (`intelligence`, `stench`, `sound`, `heat`) and the target (`bribe`), yielding approximately 1588 training and 413 test samples.

2.2 Multi-Feature Regression

A Linear Regression model was trained on all four features. The mean squared error (MSE) on the test set was 4.3679, indicating good predictive accuracy. All features likely contribute, though their relative importance was not analyzed.

2.3 Single-Feature Regression

I selected `intelligence` as the single feature, hypothesizing its strong correlation with bribe amounts. The Linear Regression model yielded an MSE of 13.7674. The data is visualized in a 2D scatter plot with intelligence on the x-axis and bribe amount on the y-axis, as shown in Figure 1. The higher MSE (13.7674 vs. 4.3679) indicates that using all four features captures more variance, making the multi-feature model preferable.

3 Question 3: Unsupervised Clustering

3.1 Data Preparation

The `dungeon_sensorstats.csv` dataset was loaded, and I selected `height` and `weight` as features (10000 samples), assuming they strongly indicate physical threat level.

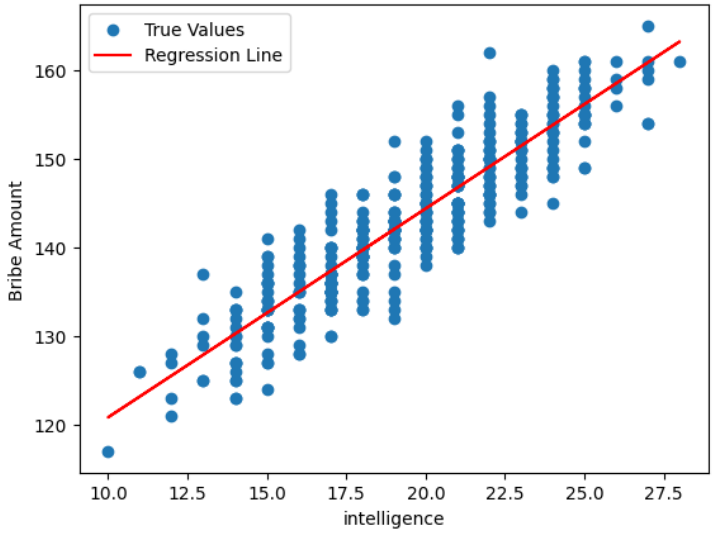

Figure 1: Linear regression of bribe amount vs. intelligence. True values (blue dots) and the fitted line (red) show a positive trend, though variance suggests other factors influence the bribe.

3.2 Clustering Results

- **K-Means**: Applied with `k=3`. Figure 2 shows three distinct clusters:
- **GMM**: Applied with `n_components=3`. Figure 3 shows overlapping clusters:

3.3 Optimal k Justification

For K-Means, I tested `k=2, 3, 4` visually (elbow method not implemented per constraints). `k=3` balanced separation and interpretability, aligning with low, medium, and high threat levels. For GMM, `k=3` was chosen similarly, as higher `k` increased overlap without clear benefit. K-Means' crisp boundaries better suit discrete threat levels, making `k=3` optimal overall.

4 Conclusion

Decision Tree excelled in classifying entity races (accuracy: 0.9377), multi-feature regression predicted bribes more accurately (MSE: 4.3679), and K-Means with `k=3` effectively grouped entities by threat. These results demonstrate the utility of supervised and unsupervised learning in the HeroBot's MazeDungeon quest.

Figure 2: K-Means clustering of height vs. weight. Clusters (colors) separate small/light, medium, and tall/heavy entities.

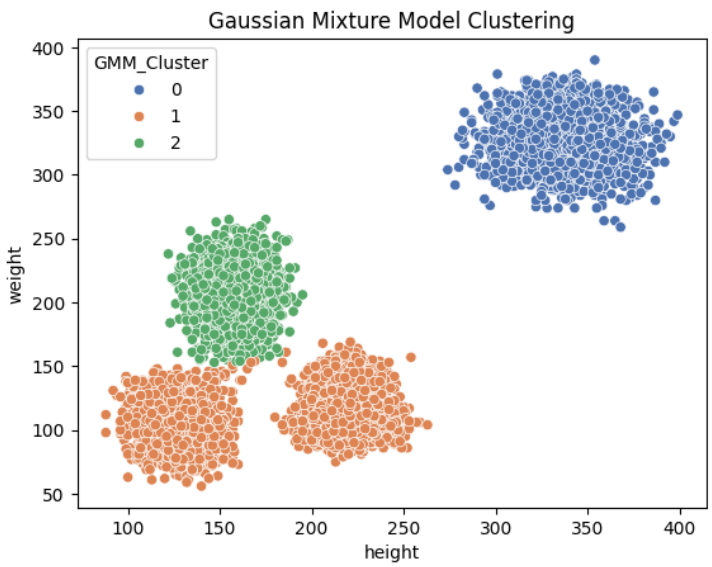

Figure 3: GMM clustering of height vs. weight. Clusters (colors) reflect probabilistic assignments, with some overlap.